Appendix G

Universality and the Finite Topology of Power

The electrodynamic isomorphism developed in this book maps social oppression onto circuit theory, thermodynamics, and control systems. The question raised in the engineering conversation is whether this isomorphism is a useful analogy or whether it reflects a deeper structural constraint: **are there only finitely many ways to organize complex systems of power and control?**

This appendix frames that question as a falsifiable conjecture.

G.1 The Universality Class Conjecture

In statistical mechanics, a **universality class** is a set of physical systems that share the same critical exponents and scaling behavior despite differing in microscopic detail. The Ising model, a lattice gas, and a binary fluid all belong to the same universality class because they share the same symmetry and dimensionality.

The conjecture is that the Predatory Min-Max Function belongs to a universality class of **hierarchical far-from-equilibrium extractors**: systems that concentrate energy, matter, or information against a gradient while maintaining stability against internal dissipation.

Conjecture G.1 (Universality of the Extraction Kernel)**.** *Any system satisfying the following three conditions will converge toward the five-node extraction topology* $(E, P_{uppet}, F_{enforce}, I_{buffer}, O_{racialized})$ *and the phase-loading control law* $\rho_k(t)$*:*

1. ***Bounded resources****: The system operates under a resource constraint that prevents uniform distribution to all nodes.*

2. ***Positive feedback****: A subset of nodes can amplify its own resource share by altering the rules of distribution.*

3. ***Dissipation threat****: The excluded nodes possess sufficient coherence capacity to threaten system stability if they mobilize.*

The conjecture claims that the *topology* of the solution is invariant: a small controlling node, an enforcement intermediary, a co-opted buffer, and an excluded sink. The specific identity axes (B_k) are culturally contingent; the circuit topology is fixed across contexts.

G.2 Evidence from Control Theory

The five-node topology is the minimal control structure capable of solving the constrained optimization problem:

$$\max_u \ \mathcal{E}(t) \quad \text{subject to} \quad M(t) < \tau,\ \dot{q} = f(q,u),\ q(0) = q_0 \tag{G.1}$$

where u is the control input and q is the state vector. Control theorists have proven that the optimal solution to such problems under resource constraints and dissipation threats takes the form of a **hierarchical feedback structure**: a leader node (E), a controller (P_{uppet}), an actuator (F_{enforce}), and a partitioned follower set ($I_{\text{buffer}} \cup O_{\text{racialized}}$) Slotine and Li 1991; Khalil 2002.

The mathematics requires a *dissipation channel*: a path through which the coherence threat $M(t)$ is converted into harmless thermal noise. The racial partition is one historically contingent instantiation of this channel. The specific choice of dissipation channel (race, caste, religion, nationality) is historically contingent. The requirement of a dissipation channel is invariant across instantiations.

G.3 Falsification Criteria

The conjecture is falsifiable if any of the following empirical conditions are met:

1. A complex society with bounded resources, positive feedback, and dissipation threat maintains stable egalitarian distribution for > 5 generations without hierarchical partition.

2. A hierarchical extractor society collapses into egalitarian equilibrium without kinetic intervention from the excluded population.

3. A system with the three conditions converges on a non-hierarchical topology (e.g., pure anarchy, rotating leadership, or stochastic resource allocation) and maintains it stably.

To date, no society in the historical dataset satisfies any of these conditions. This leaves the conjecture consistent with the dataset; it is not a proof.

G.4 Implications

If the conjecture holds, the framework's prescriptions are **engineering requirements**. The Open-Source Republic must be designed as a *different topology entirely*: one that removes the positive-feedback condition (via hard-caps on extraction) and the dissipation-threat condition (via universal baseline provisioning), thereby changing the universality class of the system.